

1 Prognostic Nutritional Index and All-Cause Mortality in
2 Gastrointestinal Cancer: A Dual-Cohort NHANES
3 Analysis With Time-Dependent, Competing Risk, and
4 Joint Inflammatory Assessment

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6 **Abstract**

7 **Background & Aims:** Nutritional status is a modifiable prognostic factor
8 in cancer patients, but population-level validation of composite nutritional in-
9 dices in gastrointestinal (GI) cancer remains limited. We aimed to validate the
10 Prognostic Nutritional Index (PNI), Controlling Nutritional Status (CONUT),
11 and Geriatric Nutritional Risk Index (GNRI) as prognostic factors for all-cause
12 mortality in GI cancer patients using nationally representative data.

13 **Methods:** We identified 313 GI cancer patients from NHANES III (1988–
14 1994, n=92) and Continuous NHANES (2005–2016, n=221) with mortality
15 follow-up through December 2019 (up to 31 years). PNI, CONUT, and GNRI
16 were derived from baseline serum albumin, lymphocyte count, total cholesterol,
17 and body mass index. Cox proportional hazards, time-dependent Cox, landmark
18 analysis, competing risks, restricted cubic splines, restricted mean survival time
19 (RMST), and population attributable fraction (PAF) analyses were performed.
20 E-value sensitivity analysis quantified robustness to unmeasured confounding.

21 **Results:** Higher PNI was associated with reduced all-cause mortality (ad-
22 justed HR 0.78, 95% CI 0.63–0.96, p=0.020). GNRI showed the strongest associ-
23 ation (HR 0.67, 95% CI 0.57–0.80, p<0.001). The PNI effect was strongly time-

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24 dependent, maximal in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$)
 25 and attenuating thereafter (interaction $p < 0.001$). Optimal PNI threshold was
 26 48.5 (cross-validated mean 48.3); patients below this had median survival 6.6
 27 vs 12.8 years ($p < 0.0001$) with RMST difference 1.97 years (95% CI 1.19–2.76,
 28 $p < 0.001$). In competing risks, PNI primarily protected against non-cancer death
 29 (HR 0.63, $p = 0.001$) rather than GI cancer-specific death (HR 1.06, $p = 0.700$).
 30 PNI decomposition showed albumin alone outperformed the composite index
 31 ($\Delta C\text{-stat} +0.032$ vs $+0.022$), while lymphocyte count contributed no indepen-
 32 dent prognostic information (HR 1.06, $p = 0.540$). Dietary quality (HEI-2015)
 33 did not predict survival (HR 0.97, $p = 0.850$). Inflammatory markers (NLR,
 34 SII) showed comparable prognostic performance to PNI. Results were consis-
 35 tent across NHANES generations (cohort interaction $p = 0.480$) and robust to
 36 adjustment for education, income, smoking, and comorbidity. Fine-Gray sub-
 37 distribution hazard models confirmed the competing risks patterns. CONUT
 38 and GNRI survived Bonferroni correction for multiple testing ($\alpha = 0.0083$),
 39 while PNI (adjusted $p = 0.020$) did not.

40 **Conclusions:** CONUT and GNRI are robust, independent prognostic fac-
 41 tors for all-cause mortality in GI cancer patients, validated across two nationally
 42 representative US cohorts spanning 30 years. PNI showed a nominally signifi-
 43 cant association that did not survive Bonferroni correction and requires inde-
 44 pendent replication. The protective effects are time-dependent, operate primar-
 45 ily through non-cancer mortality pathways, and are driven by serum albumin.
 46 These simple, inexpensive indices may guide early nutritional intervention in GI
 47 cancer patients.

48 *Keywords:* Prognostic Nutritional Index, Gastrointestinal Cancer, NHANES,
 49 Survival Analysis, Competing Risks, Time-Dependent Effect

50 1. Introduction

51 Cancer cachexia and malnutrition represent critical challenges in gastroin-
52 testinal oncology. GI cancer patients are uniquely vulnerable due to tumor-
53 induced catabolism, malabsorption, and mechanical obstruction, with malnutri-
54 tion prevalence ranging from 30% to 80% depending on tumor site and stage [1,
55 2, 3]. Malnourished patients consistently show poorer treatment tolerance, in-
56 creased postoperative complications, and reduced survival [4, 5].

57 Several composite nutritional indices have been developed to integrate multi-
58 ple biomarkers into a single prognostic score. The Prognostic Nutritional Index
59 (PNI), originally proposed by Onodera et al. in 1984, combines serum albu-
60 min and peripheral lymphocyte count [6]. The Controlling Nutritional Sta-
61 tus (CONUT) score incorporates albumin, lymphocyte count, and total cholest-
62 erol [7]. The Geriatric Nutritional Risk Index (GNRI) adjusts albumin for body
63 mass index [8]. Each index has been validated primarily in single-center surgi-
64 cal series, predominantly in East Asian populations with evidence concentrated
65 in specific GI cancer subtypes such as gastric, colorectal, and hepatocellular
66 carcinoma [9, 10, 11].

67 Despite their clinical utility, important gaps remain. First, population-level
68 validation across different geographic contexts and survey eras is lacking. Most
69 prior studies originate from single-institution cohorts with sample sizes rarely
70 exceeding 1,000 patients [12, 13, 14]. Second, the temporal dynamics of nutri-
71 tional risk—whether the prognostic effect is constant or concentrated in specific
72 post-diagnosis windows—have not been systematically examined using time-
73 dependent analyses. Third, the conceptual distinction between biochemical
74 nutritional status (serum proteins, lymphocyte count) and dietary nutritional
75 quality (nutrient intake patterns) remains poorly characterized in cancer prog-
76 nosis.

77 To address these gaps, we undertook a comprehensive analysis of nutritional
78 indices in GI cancer patients using the National Health and Nutrition Exami-
79 nation Survey (NHANES), a nationally representative population-based survey
80 with mortality linkage. We leveraged two independent NHANES generations—
81 NHANES III (1988–1994) and Continuous NHANES (2005–2016)—providing
82 up to 31 years of mortality follow-up and enabling cross-validation across differ-
83 ent survey eras. Our objectives were to: (1) validate PNI, CONUT, and GNRI
84 as prognostic factors for all-cause mortality; (2) characterize time-dependent ef-
85 fects through landmark, time-dependent Cox, and RMST analyses; (3) compare
86 prognostic performance across competing causes of death; and (4) compare bio-
87 chemical nutritional indices against a dietary quality measure (HEI-2015) and
88 inflammatory markers (NLR, SII).

89 **2. Materials and Methods**

90 *2.1. Study Population*

91 We used data from two independent NHANES cycles: NHANES III (1988–
92 1994) and Continuous NHANES (2005–2016). NHANES is a nationally repre-
93 sentative cross-sectional survey of the civilian non-institutionalized US popula-
94 tion conducted by the National Center for Health Statistics (NCHS). Both sur-
95 veys collected demographic, dietary, laboratory, and examination data through
96 standardized protocols. The analytic sample included participants aged ≥ 18
97 years with available nutrition biomarker data. A total of 44,066 participants
98 met inclusion criteria, of whom 313 had a GI cancer diagnosis.

99 The NHANES protocol was approved by the NCHS Ethics Review Board
100 (Protocol #2005-06 for 2005–2016 cycles; Protocol #98-12 and Institutional Re-
101 view Board approval for NHANES III). Written informed consent was obtained
102 from all participants. As a secondary analysis of de-identified public-use data,

103 this study was exempt from additional institutional review board approval.

104 2.2. *GI Cancer Ascertainment*

105 GI cancer was identified from self-reported cancer history. In Continuous
106 NHANES (2005–2016), cancer diagnoses were identified through the Medical
107 Conditions Questionnaire (MCQ) using cancer site codes. In NHANES III,
108 cancer site was identified from variable HAC3OS, with GI sites defined as codes
109 for colon, stomach, esophagus, pancreas, liver, and gallbladder. GI cancer was
110 defined as cancer of the esophagus, stomach, colon, rectum, pancreas, liver, or
111 gallbladder.

112 2.3. *Nutritional Indices*

113 **PNI** was calculated according to Onodera’s formula: $\text{PNI} = 10 \times \text{serum}$
114 $\text{albumin (g/dL)} + 0.005 \times \text{lymphocyte count } (\mu\text{L}^{-1})$ [6].

115 **CONUT** was derived as the sum of three component scores: albumin (≥ 3.5
116 $\text{g/dL} = 0$, $3.0\text{--}3.49 = 2$, $2.5\text{--}2.99 = 4$, $< 2.5 = 6$), lymphocyte count (≥ 1600
117 $\mu\text{L}^{-1} = 0$, $1200\text{--}1599 = 1$, $800\text{--}1199 = 2$, $< 800 = 3$), and total cholesterol (≥ 180
118 $\text{mg/dL} = 0$, $140\text{--}179 = 1$, $100\text{--}139 = 2$, $< 100 = 3$) [7].

119 **GNRI** was calculated as: $\text{GNRI} = 14.89 \times \text{albumin (g/dL)} + 41.7 \times (\text{BMI}$
120 $/ 22)$ [8].

121 Serum albumin was measured using the bromcresol purple method in Con-
122 tinuous NHANES and bromcresol green in NHANES III, with g/L values con-
123 verted to g/dL for consistency. Complete blood count provided lymphocyte
124 counts. Total cholesterol was measured enzymatically.

125 **HEI-2015** was calculated from 24-hour dietary recall data using the Food
126 Patterns Equivalents Database, measuring adherence to the 2015–2020 Dietary
127 Guidelines for Americans on a 0–100 scale.

128 2.4. Mortality Linkage

129 Mortality status was ascertained through linkage to the NCHS National
130 Death Index through December 31, 2019. Follow-up time was calculated from
131 the date of survey participation to death or censoring. The underlying cause
132 of death was classified using the UCOD_LEADING variable, with malignant
133 neoplasms (code 002) indicating cancer death.

134 2.5. Covariates

135 Demographic covariates included age (continuous, years), sex (male/female),
136 and race/ethnicity (Non-Hispanic White, Non-Hispanic Black, Hispanic, Other),
137 selected a priori based on known associations with both nutritional status and
138 mortality. Extended models additionally adjusted for education (high school
139 or above), income-to-poverty ratio, and physical activity (any moderate-to-
140 vigorous activity). Cohort (NHANES III vs. Continuous NHANES) was in-
141 cluded as a fixed effect to account for differences in albumin assay methodology
142 between survey eras.

143 2.6. Statistical Analysis

144 All analyses were conducted using R 4.6.0 with the `survey`, `survival`,
145 `EValue`, `survRM2`, and `broom` packages.

146 **Survival analysis:** Cox proportional hazards models estimated hazard ra-
147 tios (HR) per 1-standard deviation (SD) increase in each nutritional index.
148 Both crude and adjusted models (age, sex, race/ethnicity) were fitted. The pro-
149 portional hazards assumption was tested using Schoenfeld residuals. For PNI,
150 which violated this assumption ($p=0.017$), time-dependent Cox models were
151 fitted with time splitting at 2 and 5 years.

152 **Time-dependent effects:** Landmark analyses at 1, 3, and 5 years exam-
153 ined conditional survival. Restricted mean survival time (RMST) at 10 years

154 was compared between high and low PNI groups using the threshold identified
155 by maximally selected log-rank statistics [15, 16]. Restricted cubic splines (3
156 knots) examined nonlinear dose-response relationships.

157 **Competing risks:** Cause-specific Cox models were fitted for GI cancer
158 death vs. non-cancer death [17, 18]. Subdistribution hazard models (Fine-Gray)
159 were additionally fitted to address the competing risk structure from an absolute
160 risk perspective [19].

161 **Component analysis:** PNI was decomposed into albumin alone and lym-
162 phocyte alone, comparing their prognostic performance to the composite index
163 using AIC, change in C-statistic with 2000 bootstrap replications, and likelihood
164 ratio tests.

165 **Cohort-stratified and interaction analyses:** Cohort-stratified models
166 (NHANES III vs. Continuous NHANES) examined effect consistency. Formal
167 cohort-by-index interaction terms tested heterogeneity. Cohort was also in-
168 cluded as a fixed effect to address albumin assay differences (bromocresol green
169 in NHANES III vs. bromocresol purple in Continuous NHANES).

170 **Extended covariate models:** Sensitivity analyses additionally adjusted
171 for education (high school or above), income-to-poverty ratio, smoking status
172 (current vs. non-current), and a simplified Charlson comorbidity score (myocar-
173 dial infarction, congestive heart failure, stroke, chronic obstructive pulmonary
174 disease, liver disease, and diabetes) to assess robustness to socioeconomic, be-
175 havioral, and comorbidity confounding. Multiple testing was addressed using
176 Bonferroni correction for the primary family of six tests (three indices $\times$ two
177 adjustment levels).

178 **Reverse causation sensitivity:** Analyses excluding deaths within the first
179 2 years of follow-up addressed potential reverse causation from undiagnosed
180 advanced disease at baseline.

181 **Joint exposure analysis:** PNI and NLR were dichotomized at medians
182 into four groups, with additive interaction quantified using Relative Excess Risk
183 due to Interaction (RERI) with bootstrap 95% CI.

184 **Population attributable fraction (PAF):** $PAF = p \times (HR - 1) / HR$
185 was estimated for $PNI < 48.5$ with bootstrap CI [20].

186 **E-value sensitivity analysis:** E-values quantified the minimum associa-
187 tion strength an unmeasured confounder would need to fully explain away the
188 observed results [21].

189 **Bootstrap C-statistic:** Incremental discrimination (ΔC -stat) over base-
190 line covariates was estimated with 500 bootstrap replications.

191 **Inflammatory markers:** NLR and SII were compared with PNI in the
192 2005–2016 subset with available neutrophil and platelet counts.

193 3. Results

194 3.1. Baseline Characteristics

195 Of 44,066 eligible participants aged ≥ 18 , 313 (0.7%) had a GI cancer diagno-
196 sis, comprising 92 from NHANES III and 221 from Continuous NHANES. Col-
197 orectal cancer was the most common ($n=241$, 77.0%), followed by liver ($n=23$,
198 7.3%), stomach ($n=19$, 6.1%), esophageal ($n=15$, 4.8%), pancreatic ($n=8$, 2.6%),
199 and gallbladder ($n=6$, 1.9%). The remaining cases were rectal ($n=8$). GI can-
200 cer patients were substantially older than the general population (67.5 vs 45.5
201 years) and had lower PNI (50.8 vs 53.2), indicating worse nutritional status.
202 During a median follow-up of 8.8 years (up to 31 years), 169 deaths (54.0%)
203 occurred among GI cancer patients.

204 Table 1 summarizes baseline characteristics by GI tumor status, and Table 2
205 shows characteristics by PNI tertile among GI cancer patients. Patients in the
206 lowest PNI tertile had substantially higher mortality (70.5%) compared to the

middle (46.2%) and highest (45.2%) tertiles, with median survival of 6.5, 12.5, and 13.9 years, respectively (Figure 1A).

3.2. All-Cause Mortality

All three nutritional indices significantly predicted all-cause mortality after adjustment for age, sex, and race/ethnicity (Table 3). GNRI showed the strongest association (HR 0.67 per 1-SD, 95% CI 0.57–0.80, $p < 0.001$), followed by CONUT (HR 0.71, 95% CI 0.61–0.83, $p < 0.001$) and PNI (HR 0.78, 95% CI 0.63–0.96, $p = 0.020$) (Figure 2A).

Table 1: Cox proportional hazards—nutritional indices and all-cause mortality.

Index	Adjustment	HR (95% CI) per 1-SD	p-value
PNI	Crude	0.63 (0.51–0.78)	< 0.001
PNI	Adjusted*	0.78 (0.63–0.96)	0.020
CONUT	Crude	0.63 (0.55–0.73)	< 0.001
CONUT	Adjusted*	0.71 (0.61–0.83)	< 0.001
GNRI	Crude	0.67 (0.56–0.79)	< 0.001
GNRI	Adjusted*	0.67 (0.57–0.80)	< 0.001

*Adjusted for age, sex, race/ethnicity.

3.3. Time-Dependent Effects

The proportional hazards assumption was violated for PNI (Schoenfeld residuals $p = 0.017$) but not for GNRI ($p = 0.087$) or CONUT ($p = 0.430$). Time-dependent Cox models revealed marked attenuation of the PNI effect: the protective association was strongest in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$) and non-significant thereafter (2–5 year: HR 1.32, $p = 0.490$; 5+ year: HR 0.81, $p = 0.450$; interaction $p < 0.001$). GNRI showed a similar temporal pattern (0–2 year HR 0.53, $p = 0.034$), while CONUT demonstrated more stable effects across periods (Figure 3A).

In landmark analysis, PNI showed attenuated and non-significant effects at 1, 3, and 5 years, while CONUT and GNRI maintained significant associations

at all landmark times (Table 4, Figure 3B).

Table 2: Landmark analysis—adjusted HR per 1-SD.

Index	1-year	3-year	5-year
PNI	0.91 (0.73–1.13, p=0.380)	0.85 (0.66–1.10, p=0.210)	0.81 (0.60–1.09, p=0.170)
CONUT	0.75 (0.63–0.89, p<0.001)	0.71 (0.58–0.88, p=0.002)	0.74 (0.57–0.96, p=0.021)
GNRI	0.70 (0.59–0.84, p<0.001)	0.70 (0.57–0.86, p<0.001)	0.70 (0.56–0.89, p=0.003)

3.4. Optimal PNI Threshold and Dose-Response

Maximally selected log-rank statistics identified **PNI < 48.5** as the optimal prognostic threshold (p<0.0001). Patients below this threshold had markedly shorter median survival (6.6 vs 12.8 years) (Figure 1B). RMST analysis at 10 years confirmed a clinically meaningful difference of 1.97 years between groups (95% CI 1.19–2.76, p<0.001), with PAF indicating 13.4% of deaths attributable to low PNI (95% CI 6.2%–20.1%).

Restricted cubic spline analysis confirmed a nonlinear relationship between PNI and mortality risk (nonlinearity p<0.0001), with sharp risk increase below approximately 45 and a plateau above 50 (Figure 4). In contrast, CONUT and GNRI showed linear dose-response relationships (nonlinearity p=0.110 and p=0.160, respectively).

3.5. Competing Risks Analysis

Cause-specific Cox models revealed divergent patterns (Figure 2B). PNI primarily protected against non-cancer death (HR 0.63, 95% CI 0.48–0.83, p=0.001) rather than GI cancer-specific death (HR 1.06, 95% CI 0.78–1.45, p=0.700). CONUT and GNRI showed protective effects against both cancer-specific and non-cancer death (Table 5).

Fine-Gray subdistribution hazard models [19] confirmed these patterns. For GI cancer death, no index reached statistical significance (PNI sHR 1.16, 95% CI 0.77–1.75, p=0.490; CONUT sHR 0.82, 95% CI 0.59–1.14, p=0.240; GNRI

Table 3: Competing risks—cause-specific HR per 1-SD.

Index	GI Cancer Death	Non-Cancer Death
PNI	1.06 (0.78–1.45, p=0.700)	0.63 (0.48–0.83, p<0.001)
CONUT	0.73 (0.56–0.94, p=0.016)	0.70 (0.58–0.85, p<0.001)
GNRI	0.69 (0.51–0.94, p=0.017)	0.66 (0.53–0.82, p<0.001)

sHR 0.79, 95% CI 0.58–1.07, p=0.120). For non-cancer death, GNRI remained significant (sHR 0.80, 95% CI 0.64–0.99, p=0.039) and PNI was borderline (sHR 0.81, 95% CI 0.66–1.00, p=0.050).

3.6. PNI Decomposition

Albumin alone was significantly associated with mortality (HR 0.69 per 1-SD, 95% CI 0.58–0.82, p<0.001), while lymphocyte count alone was not (HR 1.06, 95% CI 0.88–1.28, p=0.540). These findings are consistent with prior systematic reviews demonstrating that low serum albumin is a robust predictor of cancer mortality across multiple tumor types [22, 23]. C-statistic improvement over baseline was +0.033 for albumin, +0.023 for PNI, and 0.000 for lymphocyte, confirming that **serum albumin is the primary driver** of PNI’s prognostic value.

3.7. PNI vs Inflammatory Markers

In the subset with neutrophil and platelet counts (n=258, 125 events), NLR (HR 1.43 per 1-SD, 95% CI 1.19–1.72, p<0.001) and SII (HR 1.30, 95% CI 1.10–1.54, p=0.002) significantly predicted mortality. These findings are consistent with prior meta-analyses demonstrating NLR as a robust prognostic marker across solid tumors [24]. C-statistic improvement was comparable across indices (+0.017 for PNI, +0.017 for logNLR). In mutually adjusted models including both PNI and logNLR, neither remained significant (PNI HR 0.86, p=0.290; logNLR HR 1.22, p=0.090), indicating shared predictive information.

269 3.8. Joint PNI $\times$ NLR Analysis

270 The “Low PNI + High NLR” group had significantly higher mortality (HR
271 1.67, 95% CI 1.06–2.63, $p=0.028$) compared to the “High PNI + Low NLR” ref-
272 erence group. RERI was -0.97 (95% CI -2.61 to -0.10), suggesting overlapping
273 rather than synergistic pathways.

274 3.9. CRP Interaction and CALLY Index

275 Among 250 patients with CRP measurements, CRP was non-significantly
276 associated with mortality (HR 1.18 per log-SD, $p=0.190$). No PNI $\times$ CRP
277 interaction was observed ($p=0.780$). The CALLY index (albumin $\times$ lymphocyte
278 / CRP) did not outperform PNI (CALLY HR 0.85, $p=0.200$; C-stat 0.659 vs
279 0.667 for PNI), despite prior studies reporting its prognostic value in colorectal
280 cancer populations [25].

281 3.10. Dietary Quality Comparison

282 Among 99 GI cancer patients with dietary data, HEI-2015 did not predict
283 mortality (HR 0.97, $p=0.850$). The correlation between HEI-2015 and PNI
284 was weak ($r=0.16$), confirming these measures capture distinct dimensions of
285 nutritional status.

286 3.11. Subgroup and Sensitivity Analyses

287 PNI showed directionally consistent protective effects across all examined
288 subgroups, with no significant effect modification by age (interaction $p=0.730$),
289 sex ($p=0.340$), or BMI ($p=0.680$) (Figure 6). Bootstrap C-statistic (500 repli-
290 cations) showed GNRI had the largest Δ C-stat ($+0.034$), followed by CONUT
291 ($+0.023$) and PNI ($+0.022$). Survey-weighted Cox (2005–2016 subset) atten-
292 uated PNI (HR 0.87, $p=0.430$) while CONUT remained significant (HR 0.76,
293 $p=0.003$). E-values ranged from 1.66 (PNI) to 1.97 (GNRI), indicating moder-
294 ate robustness to unmeasured confounding.

295 *3.12. Cohort Consistency and Extended Sensitivity Analyses*

296 Cohort-stratified analyses revealed consistent effects across NHANES gen-
297 erations. PNI showed comparable associations in NHANES III (HR 0.69, 95%
298 CI 0.51–0.93, $p=0.015$) and Continuous NHANES (HR 0.75, 95% CI 0.56–1.00,
299 $p=0.047$), with no significant cohort-by-PNI interaction ($p=0.480$). Similar con-
300 sistency was observed for CONUT (interaction $p=0.940$) and GNRI (interaction
301 $p=0.950$), supporting the appropriateness of pooling these cohorts. Inclusion of
302 cohort as a fixed effect did not materially alter the results (e.g., PNI HR 0.73,
303 95% CI 0.59–0.89).

304 Adjustment for additional socioeconomic covariates (education, income-to-
305 poverty ratio) modestly strengthened the PNI association (HR 0.74, 95% CI
306 0.55–1.00, $p=0.047$) and did not attenuate CONUT or GNRI. Further adjust-
307 ment for smoking status and Charlson comorbidity score in the Continuous
308 NHANES subset with available data ($N=133$, 61 events) did not attenuate the
309 associations: PNI HR 0.68 (95% CI 0.48–0.97, $p=0.035$), CONUT HR 0.63
310 (95% CI 0.50–0.80, $p<0.001$), and GNRI HR 0.72 (95% CI 0.55–0.93, $p=0.013$).
311 The stability of the effect estimates across progressive adjustment for socioe-
312 conomic, behavioral, and comorbidity confounders strengthens the evidence for
313 independent prognostic value, though residual confounding by unmeasured fac-
314 tors remains possible.

315 In sensitivity analyses excluding deaths within 2 years ($n=284$, 140 events),
316 PNI was no longer associated with mortality (HR 0.98, 95% CI 0.78–1.23,
317 $p=0.830$), confirming that its prognostic effect is concentrated in the early post-
318 assessment period. In contrast, CONUT and GNRI maintained significant as-
319 sociations (HR 0.77, $p=0.006$ and HR 0.70, $p<0.001$, respectively), suggesting
320 more stable long-term prognostic value.

3.13. Multiple Testing Consideration

In the primary analysis family of six tests (three indices $\times$ crude/adjusted models), the Bonferroni-corrected significance threshold was $\alpha = 0.0083$. PNI adjusted ($p=0.020$) did not survive this correction, while CONUT and GNRI adjusted (both $p<0.001$) and all crude models (all $p<0.001$) remained significant. These results are therefore reported as primary analyses for CONUT and GNRI, with PNI interpreted as a secondary finding requiring replication.

3.14. PNI Decomposition: Albumin vs. Composite Index

Formal comparison of PNI against its individual components confirmed that the prognostic value was driven entirely by serum albumin (Figure 5B). Albumin alone achieved a lower AIC (1595.8 vs. 1607.8) and larger bootstrap C-statistic improvement ($\Delta C\text{-stat} +0.032$, 95% CI 0.010–0.063) than the composite PNI ($\Delta C\text{-stat} +0.022$, 95% CI -0.001 –0.051). Lymphocyte count alone provided negligible discrimination ($\Delta C\text{-stat} +0.002$, 95% CI -0.007 –0.014). A likelihood ratio test confirmed that adding lymphocyte count to an albumin-only model did not improve fit ($p=0.460$), indicating that lymphocyte count contributes no independent prognostic information beyond albumin in this population.

4. Discussion

4.1. Summary of Principal Findings

In this population-based dual-cohort analysis spanning 30 years of NHANES data, we demonstrate that CONUT and GNRI consistently predict all-cause mortality in GI cancer patients after Bonferroni correction, while PNI showed a nominally significant but Bonferroni-non-significant association ($p=0.020$) that requires replication. GNRI showed the strongest association (HR 0.67 per 1-SD, 95% CI 0.57–0.80), followed by CONUT (HR 0.71, 95% CI 0.61–0.83) and PNI

(HR 0.78, 95% CI 0.63–0.96). The PNI effect was strongly time-dependent, maximal in the first 2 years (HR 0.34), and operated primarily through non-cancer mortality rather than GI cancer-specific pathways. Critically, the prognostic value was driven by serum albumin alone, with lymphocyte count contributing no independent information. RMST analysis quantified a 1.97-year survival difference between high and low PNI groups at 10 years.

4.2. Time-Dependent and Cause-Specific Effects

CONUT and GNRI showed stable, time-independent associations with all-cause mortality across the full follow-up period. In landmark analyses, both remained significant at 1, 3, and 5 years, and the proportional hazards assumption was not violated for CONUT ($p=0.430$) or GNRI ($p=0.087$).

The temporal variation of PNI’s prognostic effect—maximal in the first 2 years (HR 0.34) followed by marked attenuation to null after excluding early deaths—stands in contrast to the stability of CONUT and GNRI. Pre-existing nutritional status most strongly influences short-term survival, likely through its impact on treatment tolerance, perioperative recovery, and therapy completion. This pattern aligns with prior observations in a Korean gastric cancer cohort [26] and suggests that the window for effective nutritional intervention is early in the cancer trajectory. The complete attenuation after 2 years (HR 0.98 in the >2-year sensitivity analysis) also suggests that a single baseline PNI measurement cannot capture the dynamic nature of nutritional status over extended follow-up.

The competing risks analysis provides mechanistic insight: PNI primarily protects against non-cancer death (HR 0.63, $p=0.001$) rather than GI cancer-specific death (HR 1.06, $p=0.700$), suggesting PNI captures physiological reserve and resilience to treatment-related complications rather than directly influencing tumor biology. This distinction is critical for clinical nutrition: it aligns with

373 ESPEN guidelines emphasizing that nutritional therapy aims to maintain capac-
374 ity for anti-cancer treatment and reduce non-cancer complications, rather than
375 to modify cancer biology per se [1, 2]. Patients with low PNI may benefit from
376 intensified supportive care and closer monitoring for non-cancer complications.
377 CONUT and GNRI showed protective effects against both cancer-specific and
378 non-cancer death, possibly reflecting the additional prognostic information con-
379 tributed by total cholesterol and BMI, which may capture aspects of metabolic
380 health relevant to cancer progression.

381 *4.3. Nutritional vs Inflammatory Pathways*

382 PNI decomposition revealed that serum albumin alone (HR 0.69, 95% CI
383 0.58–0.82) outperformed the composite PNI (HR 0.78) across multiple metrics:
384 lower AIC (1595.8 vs. 1607.8), larger C-statistic improvement, and significantly
385 better model fit (likelihood ratio test $p < 0.001$), while lymphocyte count showed
386 no independent prognostic value ($\Delta C\text{-stat} +0.002$, 95% CI $-0.007\text{--}0.014$). This
387 finding calls into question whether the lymphocyte component of PNI adds
388 meaningful prognostic information beyond albumin alone in this population.
389 The biological rationale for PNI’s association with mortality appears to be me-
390 diated primarily by albumin’s dual role as a nutritional marker and negative
391 acute-phase reactant reflecting systemic inflammation, rather than by the im-
392 munonutritional synergy PNI was originally designed to capture.

393 In the context of current consensus frameworks, PNI and CONUT may be
394 positioned as screening tools that identify patients warranting formal GLIM
395 (Global Leadership Initiative on Malnutrition) assessment. While PNI com-
396 ponents (albumin, lymphocyte count, total cholesterol) correlate with GLIM
397 etiologic criteria (disease burden/inflammation), they do not substitute for the
398 full GLIM phenotypic assessment including weight loss, low BMI, and reduced
399 muscle mass.

400 Inflammatory markers NLR and SII showed comparable prognostic perfor-
401 mance to PNI (+0.017 C-statistic improvement for both). The mutually ad-
402 justed model, in which neither PNI nor NLR remained significant, indicates
403 substantial overlap between nutritional and inflammatory signaling pathways.
404 The joint exposure analysis (Low PNI + High NLR, HR 1.67, 95% CI 1.06–2.63;
405 RERI –0.97) suggests that combined assessment identifies the highest-risk pa-
406 tients, though the negative RERI indicates overlapping rather than synergistic
407 pathways.

408 Notably, the absence of PNI $\times$ CRP interaction ($p=0.780$) and the non-
409 significant CALLY index finding (HR 0.85, $p=0.200$) indicate that PNI’s prog-
410 nostic value extends beyond what can be explained by systemic inflammation
411 measured through single biomarkers. The null dietary finding (HEI-2015 HR
412 0.97, $p=0.850$), though limited by reliance on a single 24-hour dietary recall
413 rather than usual intake estimates, reinforces the conceptual distinction be-
414 tween biochemical nutritional status—reflecting metabolic reserve and systemic
415 inflammation—and dietary intake patterns, which may be less relevant in the
416 post-diagnosis period when metabolic alterations dominate.

417 4.4. Comparison with Previous Literature

418 Our findings extend prior single-center studies validating PNI in GI can-
419 cer. A meta-analysis of 28 studies in gastric cancer reported that low PNI was
420 significantly associated with poorer overall survival (HR 1.38, 95% CI 1.23–
421 1.54) [9], consistent with our population-level results. More recently, analyses
422 using NHANES data have similarly demonstrated PNI’s prognostic value for
423 cancer-specific and all-cause mortality in US populations [27], corroborating
424 our findings in an independent sample. A novel contribution of our study is
425 the demonstration that the PNI effect is consistent across two NHANES gen-
426 erations (cohort interaction $p=0.480$), supporting generalizability across survey

427 eras. Our study uniquely provides dual-cohort cross-validation across 30 years,
428 absolute risk estimates (RMST) that enhance clinical interpretability, and time-
429 dependent analyses demonstrating the temporal window of nutritional risk. The
430 head-to-head comparison of PNI, CONUT, and GNRI—rarely performed in a
431 single population—identifies GNRI as the most stable and strongly associated
432 index, likely because BMI adjustment captures the paradoxical protective effect
433 of preserved body mass in cancer patients. Notably, the optimal PNI thresh-
434 old of 48.5 identified in this Western population is higher than the traditional
435 cutpoint of 40–45 reported in Asian surgical series [6, 12], possibly reflecting
436 differences in body composition, comorbidity burden, and healthcare context.
437 Cross-validation (mean 48.3) confirmed this threshold was not an artifact of
438 data-driven selection, though external validation in clinical cohorts is warranted.

439 4.5. *Clinical Implications*

440 These simple, inexpensive indices can be calculated from routine laboratory
441 data in virtually any clinical setting, aligning with ESPEN recommendations for
442 routine nutritional screening in cancer patients from the time of diagnosis [1].
443 GNRI and CONUT, which incorporate BMI or total cholesterol, showed more
444 stable, linear associations across the full follow-up period and may be prefer-
445 able for long-term risk stratification when these additional measurements are
446 available. A PNI threshold of 48.5 identifies a high-risk group with substan-
447 tially shorter survival (6.6 vs 12.8 years median survival; RMST difference 1.97
448 years at 10 years), though this finding should be interpreted cautiously given
449 PNI’s Bonferroni-non-significant primary result. The concentration of risk in
450 the first 2 years after assessment suggests that early nutritional evaluation and
451 intervention may have the greatest impact, consistent with the time-dependent
452 attenuation observed for single-timepoint measurements. In the context of the
453 GLIM framework, low values of nutritional indices may trigger formal malnutri-

tion assessment, which has clearer intervention pathways. The significant PNI
× physical activity interaction ($p=0.023$) in our exploratory analysis suggests
that the association between better nutritional status and survival may be en-
hanced by physical activity, a modifiable factor warranting further investigation
in prospective studies.

4.6. *Strengths and Limitations*

Strengths include population-based sampling with national representative-
ness, dual-cohort cross-validation across 30 years, and comprehensive tempo-
ral analysis (time-dependent Cox, landmark, RCS, RMST, competing risks).
We also performed head-to-head comparison of biochemical, dietary, and in-
flammatory indices, E-value sensitivity analysis, and bootstrap C-statistic com-
parisons with absolute risk measures (RMST difference) that enhance clinical
interpretability. Extended covariate adjustment and cohort-stratified analyses
demonstrated robustness to socioeconomic confounding and cross-generational
consistency.

Several important limitations must be acknowledged. First, cancer stage,
grade, and treatment data—the strongest prognostic factors in GI cancer—are
not available in NHANES public-use data. The observed PNI-mortality associ-
ation may partly reflect confounding by disease severity, particularly given that
NHANES does not provide reliable data on time since cancer diagnosis (preva-
lent cohort design). Sensitivity analyses excluding early deaths partially address
reverse causation but cannot substitute for stage-adjusted analysis. Second,
cancer diagnoses were self-reported without medical record or cancer registry
verification, which may introduce misclassification, particularly for upper GI
sites.

Third, the use of two NHANES generations with different albumin assay
methodologies (bromocresol green in NHANES III vs. bromocresol purple in Con-

tinuous NHANES) may introduce systematic measurement differences, though cohort-stratified analyses showed consistent effects and inclusion of cohort as a fixed effect did not alter results. Fourth, nutritional indices were measured at a single baseline time point, preventing assessment of longitudinal changes in nutritional status, which may be more clinically relevant than a single snapshot. Fifth, the public-use mortality file provides only a 10-group cause-of-death recode, limiting cause-specific analysis granularity.

Sixth, the sample composition (77% colorectal cancer) limits generalizability to upper GI malignancies, and site-specific subgroup analyses were underpowered (esophageal $n=15$, pancreatic $n=8$, gallbladder $n=6$). Seventh, the pooled analysis of NHANES III and Continuous NHANES spans different treatment eras; while formal interaction tests did not identify significant heterogeneity, secular changes in GI cancer management may introduce unmeasured confounding. Eighth, adjustment for only three demographic covariates in primary models leaves potential for residual confounding. However, extended models incorporating smoking status, a simplified Charlson comorbidity score, education, and income-to-poverty ratio—applied to the subset with available smoking data—did not attenuate the associations; PNI HR remained consistent (0.68, 95% CI 0.48–0.97, $p=0.035$), and CONUT and GNRI remained robust (all $p<0.001$). These findings suggest that the PNI-mortality association is not primarily driven by smoking or comorbidity confounding, though residual confounding by unmeasured factors (alcohol use, detailed frailty measures) cannot be excluded.

Ninth, the population attributable fraction estimate (13.4%) should be interpreted with caution as it assumes a causal relationship between low PNI and mortality, an assumption that cannot be verified in an observational design. Tenth, the HEI-2015 analysis relied on a single 24-hour dietary recall, which does not represent usual dietary intake and may bias associations toward the

508 null.

509 **5. Conclusions**

510 CONUT and GNRI are independently associated with all-cause mortality
511 in GI cancer patients after Bonferroni correction, with effects consistent across
512 two NHANES generations spanning 30 years. PNI showed a nominally signifi-
513 cant association that did not survive multiple testing correction and should be
514 interpreted as a secondary finding requiring independent replication. The pro-
515 tective associations are time-dependent (particularly for PNI, concentrated in
516 the first 2 years) and operate primarily through non-cancer mortality pathways,
517 suggesting these indices capture physiological reserve and resilience rather than
518 directly reflecting cancer-specific biology. An optimal PNI threshold of 48.5
519 identifies a high-risk subgroup with a 1.97-year RMST difference at 10 years,
520 though this requires validation. The prognostic value is driven predominantly
521 by serum albumin, with lymphocyte count contributing minimal independent
522 information, raising the question of whether albumin alone may suffice for risk
523 stratification. Inflammatory markers (NLR, SII) show comparable performance,
524 and joint assessment of PNI and NLR identifies a particularly high-risk group.
525 These simple, inexpensive indices may facilitate early risk stratification and trig-
526 ger formal nutritional assessment in GI cancer patients, though future studies
527 with cancer stage, treatment data, and longitudinal nutritional measurements
528 are needed to establish their independent prognostic value.

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532 **Data Availability Statement**

533 The data used in this study are publicly available from the National Health
534 and Nutrition Examination Survey (NHANES) at [https://www.cdc.gov/nchs/
535 nhanes/](https://www.cdc.gov/nchs/nhanes/). Analysis code is available from the corresponding author upon rea-
536 sonable request.

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540 **Conflict of Interest**

541 The authors declare no conflicts of interest.

542 **Author Contributions**

543 **Zhuha Zhou:** Conceptualization, Methodology, Formal analysis, Investiga-
544 tion, Data curation, Writing—original draft, Writing—review & editing, Visu-
545 alization. **Yiqi Cai:** Supervision, Writing—review & editing. **Yongyu Bai:**
546 Investigation, Writing—review & editing. **Qigang Xu:** Data curation, Investi-
547 gation.

548 **Figure Legends**

- 549 1. **Figure 1:** Kaplan-Meier survival curves. (A) By PNI tertile (low, mid,
550 high); (B) By PNI threshold 48.5, with median survival and p-values.
- 551 2. **Figure 2:** Forest plots of adjusted HR per 1-SD. (A) All-cause mortality
552 (Cox); (B) Competing risks: cancer death vs non-cancer death (Fine-
553 Gray).

- 554 3. **Figure 3:** Time-dependent analysis. (A) Hazard ratios across follow-up
555 periods (0–2, 2–5, 5+ years); (B) Landmark analysis at 1, 3, and 5 years.
- 556 4. **Figure 4:** Restricted cubic spline dose-response curve for PNI (3 knots),
557 with 95% CI.
- 558 5. **Figure 5:** Δ C-statistic comparison. (A) PNI vs inflammatory markers
559 (NLR, SII); (B) PNI decomposition: albumin, PNI, and lymphocyte.
- 560 6. **Figure 6:** Subgroup forest plot—PNI effect across demographic and clin-
561 ical strata.

562 **Supplementary Materials**

563 **Tables S1–S7** and **Figures S1–S3** are available online as supplementary
564 data.

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